

RemoTeC_create and RemoTeC_sim v2.3.7 – Manual

Date: 19/12/2014

Authors: André Butz (andre.butz@kit.edu) & Haili Hu (h.hu@sron.nl)

Obtaining the program

The simulation package of RemoTeC has a private repository on Bitbucket. If you are not a RemoTeC team member but wish to obtain the code, please contact André Butz or Haili Hu. Team members can obtain the code with:

```
> hg clone https://username@bitbucket.org/remotec/sim
```

You will be asked for your password of your Bitbucket account four times because there are in total four repositories (sim, sim_retrieval, sim_create and remotec_core).

Compiling the program

The simulation package consists of two executables, `RemoTeC_create` and `RemoTeC_sim`, one for the creation of synthetic measurements and one for the retrieval from synthetic measurements, respectively.

The programs `RemoTeC_create` and `RemoTeC_sim` are compiled in normal (processing, sequential) mode with:

```
> make [RemoTeC_create] [RemoTeC_sim]
```

if no `RemoTeC_create` or `RemoTeC_sim` arguments are given, both executables are built.

Compilation in debugging mode is with:

```
> make debug=yes
```

Compilation in parallel mode (using OpenMP) is with:

```
> make parallel=yes
```

On SRON platforms, the build process has been tested with the ifort and gfortran compiler; on KIT platforms with the sun compiler:

```
> make FORT=SRON-ifort
> make FORT=SRON-gfortran
> make FORT=KIT-sun
```

Launching the program Remotec_create

The program `RemoTeC_create` is launched from the command line via:

```
> ./RemoTeC_create Run-ID DB-File [Run-Path]
```

For example:

```
> ./RemoTeC_create 1 INI/test_atm /home/andreb/
```

Run-ID is a required integer between 0 and 9999 that identifies the running instance of the program. The program uses Run-ID to generate the output filenames. This is convenient for running several instances of the program in the same folder without causing conflicts when writing the output. Furthermore, configuration settings are read in from `"Run-Path/INI/settingsRun-ID.nml"`, thus allowing different runs to have different settings.

DB-File is a required character array that points to a file which contains a list of atmospheric scenario files to be submitted to the program. Each line in the DB-File must contain the name of an atmospheric scenario file. Different formats for the atmospheric scenario file are possible, this can be selected in the configuration file `"settingsRun-ID.nml"` (see section **Required input: INI/settingsRun-ID.nml**).

[Run-Path] is an optional character array which contains the directory in which the program is to

be executed. The settings inputfile is read from "Run-Path/INI/." If no Run-Path command line argument is given, the current directory (".") is used.

The above example uses "1" as Run-ID, "INI/test_atm" as DB-File and "/home/andreb/" as Run-Path. Thus, the program will be run in "/home/andreb/"; configuration settings are read from "/home/andreb/INI/settitngs0001.nml"; the atmospheric scenarios for which spectra are to be generated are listed in "/home/andreb/INI/test_atm".

Launching the program Remotec_sim

The program RemoTeC_sim is launched from the command line via:

```
> ./RemoTeC_sim Run-ID DB-File [Run-Path]
```

For example:

```
> ./RemoTeC_sim 2 INI/test_spectra /home/andreb/
```

Run-ID is a required integer between 0 and 9999 that identifies the running instance of the program. The program uses Run-ID to generate the output filenames. This is convenient for running several instances of the program in the same folder without causing conflicts when writing the output. Furthermore, configuration settings are read in from "Run-Path/INI/settingsRun-ID.nml", thus allowing different runs to have different settings.

DB-File is a required character array that points to a file which contains a list of synthetic spectra to be submitted to the program. Each line in the DB-File must contain the basename of a synthetic spectra file. The code assumes that the synthetic spectra file is named "L1B_basename" and, the meteo data file is named "METEO_basename" or "ATM_basename", depending on which format is specified in the the configuration file "settingsRun-ID.nml" (see section **Required input: INI/settingsRun-ID.nml**).

[Run-Path] is an optional character array which contains the directory in which the program is to be executed. The settings inputfile is read from "Run-Path/INI/." If no Run-Path command line argument is given, the current directory (".") is used.

The above example uses "2" as Run-ID, "INI/test_spectra" as DB-File and "/home/andreb/" as Run-Path. Thus, the program will be run in "/home/andreb/"; configuration settings are read from "/home/andreb/INI/settitngs0002.nml"; the synthetic measurements to be analyzed are listed in "/home/andreb/INI/test_spectra".

Required input: INI/settingsRun-ID.nml

RemoTeC_create and RemoTeC_sim require a configuration file in the form of a namelist inputfile called "Run-Path/INI/settingsRun-ID.nml". It defines required input variables and options how to run the program. Appendix A and B list an example file in its entirety for RemoTeC_create and RemoTeC_sim, respectively. Below the example is explained per namelist. Some input variables have a different function, depending whether it is used by RemoTeC_create or RemoTeC_sim. This will be explained if needed.

&paths

```
path%spectrum = 'TEST/SYNTH_SPECTRA/'
```

A character array that defines the parent path to the synthetic measurements. For RemoTeC_create this is the directory where the synthetic measurements will be written to. For RemoTeC_sim this is the directory where the synthetic L1B spectra are assumed to be located.

```
path%meteo = 'TEST/SYNTH_SPECTRA/'
```

A character array that defines the parent path to the atmospheric scenarios for RemoTeC_create and meteo data for RemoTeC_sim.

```
path%ils=
```

```
'/stage/nobackup/haili/Tropomi_input/ISRF/S5P_TEST_AUX_ISRF___00000000T0000000_99999999T9999999_20141126T125935.nc'
```

A character array that defines the path to the instrument line shape file, only relevant if ILS is read from file (see flag%ilscal).

```
path%sun=
```

```
'/stage/nobackup/haili/Tropomi_input/SUN/S5P_TEST_REF_SOLAR___00000000T0000000_99999999T9999999_20141126T130457.nc'
```

A character array that defines the path to the model solar spectra.

```
path%mie = '/GOSAT/AUX/MIE/aerosol_lut.nc'
```

A character array that defines the path to the look-up-table of aerosol optical properties generated by Mie calculations.

```
path%cirrus = '/GOSAT/AUX/CIRRUS/cirrus_lut.nc'
```

A character array that defines the path to the look-up-table of cirrus optical properties generated by raytracing calculations.

```
path%output = 'CONTRL_OUT/' /
```

A character array that defines the path where the outputfiles are written to: e.g. CONTRL_OUT/statistics0002.dat for RemoTeC_sim.

&flags

```
flag%atm = 2
```

An integer that specifies the format of the atmospheric scenario for RemoTeC_create (0=custom; 1=AFGL; 2=ECHAM, 3=SICOR). For RemoTeC_sim, it specifies the format of the meteo datafile (1=ATM_??; 2=METEO_??). ATM_?? Contains the meteo data on the same pressure grid as the input atmospheric scenario used by RemoTeC_create, METEO_?? contains the meteo data on the 91 ECMWF's hybrid levels.

```
flag%scat = 1
```

An integer that selects the non-scattering (0) or scattering (1) forward model. For option (1), the aerosol type(s) should be specified further on in namelist(s) &aerosol??. For RemoTeC_create, there are extra options: (2) aerosol and cirrus from model/measurements; (3) aerosol from model/measurements (no cirrus); (4) cirrus from model/measurements (no aerosol).

```
flag%rtm = 1
```

An integer that selects which RTM to use: LINTRAN 1.0 (1), LINTRAN 2.1 with Delta-M correction (2), LINTRAN 2.1 with Nakajima correction (3). Option (1) is default, options (2) and (3) are being tested.

```
flag%inv = 3
```

Only relevant for RemoTeC_sim. An integer that selects the inverse method used: truncated SVD (0), 0th order Phillips-Tikhonov regularization (1), or an ad-hoc 1st order Phillips-Tikhonov (3). If the number of retrieval layers equals 1, this flag is not used and a mere least-squares fit is performed. Option (2) is for profile scaling with regularization of aerosol parameters and is currently being tested.

```
flag%ils = 2
```

An integer that selects the method of ILS convolution: standard (1) or FFT (2).

```
flag%ilscal = 0
```

An integer that specifies whether the ILS is read from file (0=from S5P file; 2=from custom file) or calculated inline with a Gaussian (1).

```
flag%fit = 2
```

Only relevant for RemoTeC_sim. An integer that defines whether to fit Reflectance (1) or Radiance (2).

```
flag%temp = 0
```

Only relevant for RemoTeC_sim. An integer that defines whether to fit (1) or not to fit (0) an offset of the temperature profile.

```
flag%Fs = 0
```

For RemoTeC_create: An integer that defines whether to include fluorescence emission as a constant (1) or two-added Gaussians (2) or not at all (0). The relevant fluorescence parameters are currently hardcoded and may need to be revised.

For RemoTeC_sim: An integer that defines whether to fit (1) or not to fit (0) fluorescence emission. The current implementation is based on Eq. (4) of Frankenberg et al. (2012) and only works for flag%scat=1, it is being tested.

```
flag%xs = 0
```

An integer that specifies whether to read the entire cross section databases into memory (<1) or whether to read only a specific set of cross sections (>0) from CONTRL_OUT/cross_section_?.dat. For flag%xs<0, the cross sections are written to CONTRL_OUT/cross_section_?.dat. Default should be flag%xs=0, the other options are used for development purposes. If a single spectrum is treated over and over again, there is no use in reading all the cross section databases into memory over and over again. It is much more convenient and faster to just read the same set of cross sections from ASCII.

```
flag%output = 0 /
```

An integer that selects what output is to be generated: 0=standard output; 1=0+faulty spectra/state vector; 2=1+screen output; 3=2+debug output.

&alt_grid

The 3 integers below define the atmospheric height grids used by the program. The below example uses 12 retrieval layers, with 3 RT sublayers each. Each RT sublayer has 2 sublayers for the cross section calculation. Thus, vertical profiles are retrieved in 12 atmospheric layers; the RT is calculated on 36 layers; the cross sections are calculated for 72 layers.

```
grid%nlay = 12
```

An integer that defines the number of retrieval layers

```
grid%nrt = 3
```

An integer that defines the sublayers for the radiative transfer (RT) calculations. The total number of layers in RTM = nlay*nrt.

```
grid%natm = 2 /
```

An integer that defines the sublayers for the absorption cross-section calculation. The total number of layers for cross-sections = nlay*nrt*natm.

&dim

nwin = 2

An integer that defines the number of spectral windows. Up to 10 windows are in principle possible. The example considers 2 windows. This implies that "INI/settingsRun-ID.nml" must contain 2 times the window input structure: &window1 and &window2.

ntype_aer = 1

An integer that defines the number of particle types. Up to 10 types are in principle possible. The example considers 1 aerosol type. This implies that "INI/settingsRun-ID.nml" must contain 1 times the particle input structure: &aerosol1.

&nabs

ntype_abs = 1, 3 /

A set of integers that defines the number of considered absorbers for the different retrieval windows.

&window1

wave_start = 12920.

A float number that gives the starting point of the window in wavenumber units (1/cm).

wave_stop = 13195.

A float number that gives the ending point of the window in wavenumber units (1/cm).

reso = .01

A float number that defines the sampling in wavenumber units (1/cm) to be used for the forward model calculations.

fwhm = 6.48

A float number that defines the full-width-at-half-maximum of the instrument function if the ILS is to be calculated inline (flag%ilscalcul=1). fwhm is also used to calculate the user-defined number of ILS points per spectral pixel, this applies also when the ILS is read from file.

samp = 5.

Only relevant for RemoTeC_create. A float number that defines the oversampling ratio: the spectral sampling interval is defined by $fwhm/samp$.

wvbd = 120.

A float number that defines the width the model spectral range is to be extended with in units of fwhm. The forward model calculations consider a wider spectral range than the range defined by wave_start and wave_stop. The spectral range is extended with $wvbd*fwhm$ at both ends. This accounts for the fact that spectral points outside the actual retrieval window contribute to the signal within the retrieval window due to the finite extent of the instrument function,

ntau = 20

ntau_2nd_max = 1

Two integers that define the number of bins for the efficient linear-k binning RT for the primary absorber and the secondary absorbers, respectively. If ntau=0, then line-by-line RT is used. If there is only 1 absorber in the window ntau_2nd_max is automatically set to 1.

type_x(1) = -7

An integer that defines the type of the absorber following the HITRAN notation (e.g. 1=H2O, 2=CO2, 5=CO, 6=CH4, 7=O2).

For RemoTeC_sim:

Note that absorbers of the same type are linked among retrieval windows. If the same absorber e.g. 2=CO2 is present in several windows, the retrieval tries to find the concentration that

simultaneously fits best all the respective windows. There is the option to unlink the absorbers by artificially changing the type of the absorbers. For CO2 for example, the absorbers 200 to 299 are treated as if they were CO2 but there is no link to absorber type 2. In analogy this applies to all absorber types (x, x00 to x99).

Note that there is the option to consider an absorber in a spectral window but not to retrieve its concentration by setting the absorber type a negative number. If for example oxygen absorption is present in the considered retrieval window, but there is no fitting required because its concentration is a priori known, it can be initialized as absorber type -7. In analogy this applies to all absorber types.

Note that oxygen (absorber type 7) is treated different from the other absorbers. While the program in general assumes the air column (derived from pressure) fixed and the absorber fraction variable, oxygen retrievals assume the fraction fixed (=20.95%) and the air column variable. If the retrieval finds an oxygen column which differs from the initial guess, also pressure and air column are adjusted. If this is not desired, there is the option to initialize oxygen as absorber type 700 to 799 which are treated as regular absorbers with fixed air column.

For RemoTeC_create:

7, -7 or 700 to 799 have the same effect.

type_x_flag(1) = 0

Only relevant for RemoTeC_sim. An integer that specifies whether the absorber total column (0) or the absorber profile (1) is to be retrieved.

type_xsdb(1) = 2

An integer that specifies the format of the cross section database: Frankenberg (0), Butz (1), Galli NetCDF (2) or S5P NetCDF (3).

xsdb_path(1)=

'/stage/nobackup/haili/Tropomi_input/XSDB/HITRAN12/hit08_07_LM_LmCia_12800.00-13300.00_spec_res_0.005.nc'

A character array that defines the path to the cross section database.

albflag = 3

For RemoTeC_sim:

An integer that defines the number of parameters fitted for surface albedo. If N parameters are considered, a polynomial of degree N-1 is fitted.

For RemoTeC_create:

An integer that defines which albedo to use: 0=use satellite data , 1=0th order, 2=1st order, 3=2nd order polynomial.

albedo = 0. 0. 0.

Only relevant for RemoTeC_create and albflag>0. A set of float numbers that define the albedo in polynomial form up to order albflag-1.

IoffFlag = 1

Only relevant for RemoTeC_sim. An integer that defines the number of parameters fitted for additive intensity offset. If N parameters are considered, a polynomial of degree N-1 is fitted.

spsh0flag = 1

Only relevant for RemoTeC_sim. An integer that defines whether a shift of the spectral grid is fitted in units of wavenumbers (-1) or wavelength (1) or not (0).

spsh1flag = 0

Only relevant for RemoTeC_sim. An integer that defines whether a first order stretch/squeeze of the spectral grid is fitted in units of wavenumbers (1) or wavelength (-1) or not (0).

spsh2flag = 0

Only relevant for `RemoTeC_sim`. An integer that defines whether a second order stretch/squeeze of the spectral grid is fitted in units of wavenumbers (1) or wavelength (-1) or not (0).

`sunsh0flag = 1 /`

Only relevant for `RemoTeC_sim`. An integer that defines whether a shift of the solar spectra (high resolution) is fitted in units of wavenumber (1) or wavelength (-1) or not (0).

&aerosol1

`CirrusFlag = 0`

An integer that specifies whether the particles are Mie-type aerosols (0) or randomly oriented hexagonal cirrus particles (1).

`amount_fit = 1`

Only relevant for `RemoTeC_sim`. An integer that specifies whether to fit (1) or not to fit (0) the total column.

`tau_ref = 0.1`

A float number that defines (the initial guess for) particle optical thickness at 765 nm.

`height_dist = 0`

An integer that defines the type of height distribution: Gauss layer (0), exponential decay (1), box (2), Lorentz layer (3), Gauss layer with variable width (4).

`alt1_fit = 1`

Only relevant for `RemoTeC_sim`. An integer that specifies whether to fit (1) or not to fit (0) the first height distribution parameter.

`alt1 = 5000`

A float number that defines (the initial guess for) the first height distribution parameter. This parameter is the center of the particle layer for the Gauss, box, and Lorentz distributions. For the exponential distribution the respective parameter is the height of the boundary layer where the particle optical thickness is assumed uniform.

Note that the first height distribution parameter cannot be fitted for the exponential and the box distribution.

`alt2_fit = 0`

Only relevant for `RemoTeC_sim`. An integer that specifies whether to fit (1) or not to fit (0) the second height distribution parameter.

`alt2 = 2000`

A float number that defines (the initial guess for) the second height distribution parameter. This parameter is the width of the particle layer for the Gauss, box, and Lorentz distributions. For the exponential distribution the respective parameter is the exponent by which the particle optical thickness decays (as a function of pressure) above the boundary layer. For the variable-width-Gauss distribution this parameter defines how the layer width depends on layer height.

Note that the second height distribution parameter cannot be fitted for the variable-width-Gauss and the box distribution.

`rm_fit = 0`

Only relevant for `RemoTeC_sim`. An integer that specifies whether to fit (1) or not to fit (0) the real part of the particle refractive index.

`rm(:) = 1.4 1.47`

A set of float numbers that defines the initial guess of real part of refractive index for each window. Note that if the refractive index is not fitted, the value may differ among the considered

retrieval windows. If refractive index is fitted, the retrieval fits a single global refractive index to all windows.

`fim_fit = 0`

Only relevant for `RemoTeC_sim`. An integer specifying whether to fit (1) or not to fit (0) the imaginary part of the particle refractive index.

`fim(:) = -0.01 -0.008`

Set of float numbers that define the initial guess of imaginary part of refractive index for each window. Note that if the refractive index is not fitted, the value may differ among the considered retrieval windows. If refractive index is fitted, the retrieval fits a single global refractive index to all windows.

`size_dist = 0`

An integer that specifies the type of particle size distribution: power law (0), bimodal lognormal (1), or gamma (2).

`reff_fit = 1`

Only relevant for `RemoTeC_sim`. An integer that specifies whether to fit (1) or not to fit (0) the first size distribution parameter. For the power law size distribution, this is the power parameter, for the other size distributions it is the effective radius of the fine mode.

`reff = 3.5`

A float number that defines the initial guess of the first size distribution parameter.

`veff_fit = 0`

Only relevant for `RemoTeC_sim`. An integer that specifies whether to fit (1) or not to fit (0) the second size distribution parameter. For the bimodal and the gamma distribution this is the effective width of the fine mode. For the power law size distribution, this is the upper boundary of the size distribution in the case of Mie-type aerosols. For cirrus particles and power law, this parameter is overwritten. This parameter cannot be fitted for the power law.

`veff = 10.`

A float number that defines (the initial guess of) the second size distribution parameter.

`shapefrac = 1.`

A float number that defines the fraction of spheres to spheroids in the case of Mie-type aerosols. In the case of cirrus particles, this parameter defines the fraction of columnar to plate-like ice crystals.

`shapefrac_fit = 0 /`

Only relevant for `RemoTeC_sim`. An integer that specifies whether to fit (1) or not to fit (0) the shape fraction.

Appendix A: settingsRun-ID.nml for RemoTeC_create

```
!*****
!*** Namelist input for creating synthetic measurements
!*** atmosphere and aerosols from ECHAM
!*****

!*** Paths of input/outputfiles
&paths
  path%spectrum = 'SYNTH_SPECTRA/'
  path%meteo = 'TEST/ATM_INPUT/'
  path%sun
    =
    '/stage/nobackup/haili/Tropomi_input/SUN/S5P_TEST_REF_SOLAR___00000000T000000_999999
    99T999999_20141126T130457.nc'
  path%ils
    =
    '/stage/nobackup/haili/Tropomi_input/ISRF/S5P_TEST_AUX_ISRF___00000000T000000_99999
    99T999999_20141126T125935.nc'
  path%mie = '/stage/nobackup/haili/Tropomi_input/Oleg_DB_new/aerosol_lut.nc'
  path%cirrus = '/stage/nobackup/haili/Tropomi_input/CIRRUS_TABLE/cirrus_lut.nc'
  path%output = 'CONTRL_OUT/' /

&flags
!*** atmosphere:
!*** 0=custom
!*** 1=AFGL
!*** 2=ECHAM
!*** 3=SICOR
  flag%atm = 2
!*** Scattering:
!*** 0=non-scattering,
!*** 1=scattering with aerosol and cirrus specified below,
!*** 2=aerosol and cirrus from model/measurements
!*** 3=aerosol from model/measurements (no cirrus)
!*** 4=cirrus from model/measurements (no aerosol),
  flag%scat = 2
!*** RTM: 1=LINTRAN 1.0(Nakajima), 2=LINTRAN 2.1 (Delta-M), 3=LINTRAN2.1 (Nakajima)
  flag%rtm = 1
!*** fluorescence emission
!*** 0=none
!*** 1=constant
!*** 2=two co-added Gaussians
  flag%Fs=0
!*** ILS convolution: 1=standard convolution, 2=Fast Fourier Convolution
  flag%ils = 1
!*** ISRF: 0=from S5P file, 1=inline calculation of Gaussian ISRF, 2=from custom
file
  flag%ilscal = 0
!*** Preprocessing of absorber cross-sections (regrid to internal spectral grid):
!*** 1=linear interpolation, 2=triangular convolution
  flag%xs_preprocess = 1
!*** Output: 0=standard output, 1=0+extra output, 2=1+screen output
  flag%output = 2 /

&alt_grid
!*** flag: 1=equidistant pressure grid, 2=equidistant altitude grid
  grid%flag = 1
```

```

!*** number of retrieval layers
    grid%nlay = 12
!*** number of layers in RTM = nlay*nrt
    grid%nrt = 3
!*** number of layers for cross-sections = nlay*nrt*natm
    grid%natm = 2 /

&dim
!*** number of spectral windows
    nwin = 2 /

&nabs
!*** number of absorbers per window
    ntype_abs(:) = 1, 3 /

&window1
!*** lower boundary of spectral window [1/cm]
    wave_start = 12920.
!*** upper boundary of spectral window [1/cm]
    wave_stop = 13195.
!*** sampling
    reso = 0.1
!*** full-width-half-maximum
    fwhm = 6.48
!*** Oversampling ratio
    samp = 5.
!*** wavenumber boundary offset (in multiples of fwhm)
    wvbd = 5.
!*** Pair of bins for RT K-binning (#1st absorber > #2nd absorber)
    ntau = 10
    ntau_2nd_max = 1
!*** Absorber type (HITRAN index)
    type_x(1) = -7
!*** Targetflag: 0=interfering absorber, 1=target absorber
    type_x_flag(1) = 0
!*** Format of cross-section database: 0=Frankenberg, 1=Butz, 2=netCDF
    type_xsdb(1) = 2
!*** Path to XSDB
    xsdb_path(1)
' /stage/nobackup/haili/Tropomi_input/XSDB/HITRAN12/hit08_07_LM_LmCia_12800.00-
13300.00_spec_res_0.005.nc'
!*** Albedo: 0=use satellite data , 1=0th order, 2=1st order, 3=2nd order
    albflag = 0 /
!*** If albflag > 0, set albedo below
!    albedo = 0. 0. 0./

&window2
    wave_start = 4190.
    wave_stop = 4340.
    reso = 0.02
    fwhm = 0.45
    samp = 2.5
    wvbd = 5.
    ntau = 5
    ntau_2nd_max = 4

```

```

type_x(1) = 6
type_x_flag(1) = 1
type_xsdb(1) = 2
xsdb_path(1)
'/stage/nobackup/haili/Tropomi_input/XSDB/HITRAN12/hit12_06_Voigt_4150.00-
4370.00_spec_res_0.005.nc'
type_x(2) = 1
type_x_flag(2) = 0
type_xsdb(2) = 2
xsdb_path(2)
'/stage/nobackup/haili/Tropomi_input/XSDB/HITRAN12/hit12_01_Voigt_4150.00-
4370.00_spec_res_0.005.nc'
type_x(3) = 5
type_x_flag(3) = 0
type_xsdb(3) = 2
xsdb_path(3)
'/stage/nobackup/haili/Tropomi_input/XSDB/HITRAN12/hit12_05_Voigt_4150.00-
4370.00_spec_res_0.005.nc'
albflag = 0/

```

Appendix B: settingsRun-ID.nml for RemoTeC_sim

```

!*****
!*** Namelist input for retrieval of synthetic measurements
!*** 1 aerosol type
!*****

!*** Paths of input/outputfiles
&paths
  path%spectrum = 'TEST/SYNTH_SPECTRA/'
  path%meteo = 'TEST/SYNTH_SPECTRA/'
path%ils
'/stage/nobackup/haili/Tropomi_input/ISRF/S5P_TEST_AUX_ISRF___00000000T000000_99999
999T999999_20141126T125935.nc'
  path%sun
'/stage/nobackup/haili/Tropomi_input/SUN/S5P_TEST_REF_SOLAR___00000000T000000_999999
99T999999_20141126T130457.nc'
  path%mie = '/stage/nobackup/haili/Tropomi_input/Oleg_DB_new/aerosol_lut.nc'
  path%cirrus = '/stage/nobackup/haili/Tropomi_input/CIRRUS_TABLE/cirrus_lut.nc'
  path%output = 'CONTRL_OUT/' /

&flags
!*** Scattering: 0=non-scattering, 1=scattering
  flag%scat = 1
!*** RTM: 1=LINTRAN 1.0(Nakajima), 2=LINTRAN 2.1 (Delta-M), 3=LINTRAN2.1 (Nakajima)
  flag%rtm = 1
!*** atmosphere 1=ATM, 2=METEO
  flag%atm = 2
!***Inversion: 0=TSVD, 1=PHILLIPS-TIKHONOV, 3=ADHOC
  flag%inv = 1
!*** Fit temperature offset, 0=no, 1=yes
  flag%temp = 0
!*** Fit fluorescence, 0=no, 1=yes
  flag%Fs = 0
!*** ILS convolution: 1=standard convolution, 2=Fast Fourier Convolution
  flag%ils = 1

```

```

!*** ISRF: 0=from S5P file, 1=inline calculation of Gaussian ISRF, 2=from custom
file
    flag%ilscal = 0
!*** Fit spectrum with: 1=reflectance, 2=radiance
    flag%fit = 2
!*** Preprocessing of absorber cross-sections (regrid to internal spectral grid):
!*** 1=linear interpolation, 2=triangular convolution, 3=rectangular convolution
    flag%xs_preprocess = 1
!*** Output: 0=standard output, 1=0+state vector, 2=1+screen output, 3=2+debug
output
!*** WARNING: Do not use flag%output=3 for parallel runs
    flag%output = 2 /

&alt_grid
!*** flag: 1=equidistant pressure grid, 2=equidistant altitude grid
    grid%flag = 1
!*** number of retrieval layers
    grid%nlay = 12
!*** number of layers in RTM = nlay*nrt
    grid%nrt = 3
!*** number of layers for cross-sections = nlay*nrt*natm
    grid%natm = 2 /

&dim
!*** number of spectral windows
    nwin = 2
!*** number of aerosol types (including cirrus)
    ntype_aer = 1 /

&nabs
!*** number of absorbers per window
    ntype_abs(:) = 1, 3 /

&window1
!*** lower boundary of spectral window [1/cm]
    wave_start = 12920.
!*** upper boundary of spectral window [1/cm]
    wave_stop = 13195.
!*** sampling
    reso = 0.1
!*** full-width-half-maximum
    fwhm = 6.48
!*** Oversampling ratio
    samp = 5.
!*** wavenumber boundary offset (in multiples of fwhm)
    wvbd = 5.
!***Pair of bins for RT K-binning(#1st absorber > #2nd absorber)
    ntau = 10
    ntau_2nd_max = 1
!*** Absorber type (HITRAN index)
    type_x(1) = -7
!*** Targetflag: 0=interfering absorber, 1=target absorber
    type_x_flag(1) = 0
!*** Format of cross-section database: 0=Frankenberg, 1=Butz, 2=netCDF (XSDB-
format), 3=netCDF (S5P-format)

```

```

    type_xsdb(1) = 3
!*** Path to XSDB
    xsdb_path(1) =
'/stage/nobackup/haili/Tropomi_input/XSDB/S5P_OPER_REF_XS_CH4_00000000T000000_99999
999T999999_20140908T115152.nc'
!*** Fit albedo: 0=no, 1=constant, 2=1st order spectral dependence, 3=2nd order
spectral dependence
    albflag = 2
!*** Fit Intensity offset: 0=no, 1=yes
    IoffFlag = 0
!*** Fit 0th order spectral shift: 0=no, 1=wavenumber shift, -1=wavelength shift
    spsh0flag = -1
!*** Fit 1st order spectral shift: 0=no, 1=wavenumber shift, -1=wavelength shift
    spsh1flag = 0
!*** Fit 2nd order spectral shift: 0=no, 1=wavenumber shift, -1=wavelength shift
    spsh2flag = 0
!*** Fit 0th order spectral shift of sun: 0=no, 1=wavenumber shift, -1=wavelength
shift
    sunsh0flag = 0 /

&window2
    wave_start = 4190.
    wave_stop = 4340.
    reso = .02
    fwhm = 0.45
    samp = 2.5
    wvbd = 5.
    ntau = 5
    ntau_2nd_max = 4
    type_x(1) = 6
    type_x_flag(1) = 1
    type_xsdb(1) = 3
    xsdb_path(1) =
'/stage/nobackup/haili/Tropomi_input/XSDB/S5P_OPER_REF_XS_CH4_00000000T000000_99999
999T999999_20140908T115152.nc'
    type_x(2) = 1
    type_x_flag(2) = 0
    type_xsdb(2) = 3
    xsdb_path(2) =
'/stage/nobackup/haili/Tropomi_input/XSDB/S5P_OPER_REF_XS_CH4_00000000T000000_99999
999T999999_20140908T115152.nc'
    type_x(3) = 5
    type_x_flag(3) = 0
    type_xsdb(3) = 3
    xsdb_path(3) =
'/stage/nobackup/haili/Tropomi_input/XSDB/S5P_OPER_REF_XS_CH4_00000000T000000_99999
999T999999_20140908T115152.nc'
    albflag = 2
    IoffFlag = 0
    spsh0flag = -1
    spsh1flag = 0
    spsh2flag = 0
    sunsh0flag = 0 /

&aerosol1
!*** Cirrus: 0=no, 1=yes
    CirrusFlag = 0

```

```

!*** Fit amount: 0=no, 1=yes
    amount_fit = 1
!*** (Initial) aerosol optical thickness
    tau_ref = 0.1
!*** Height distribution:
!*** 0=GAUSS, 1=EXPONENTIAL, 2=BOX, 3=LORENTZ, 4=GAUSS(1-PAR),
!*** 5=DOUBLE-GAUSS, 6=DOUBLE BOX, 7=ONE-LAYER, 8=GAUSS(over restricted altitude
range)
    height_dist = 0
!*** Fit height distribution alt1: 0=no, 1=yes
    alt1_fit = 1
!*** (Initial) height distribution parameter alt1
!*** if alt1<0, then alt1 is set to height of boundary layer for aerosol and
tropopause for cirrus)
    alt1 = 5000
!*** Fit height distribution parameter alt2: 0=no, 1=yes
    alt2_fit = 0
!*** (Initial) height distribution parameter alt2
    alt2 = 2000
!*** Fit real refractive index: 0=no, 1=yes
    rm_fit = 0
!*** (Initial) real refractive index per window
    rm(:) = 1.4 1.47
!*** Fit imaginary refractive index: 0=no, 1=yes
    fim_fit = 0
!*** (Initial) imaginary refractive index per window
    fim(:) = -0.01 -0.008
!*** Size distribution: 0=POWER LAW, 1=LOGNORMAL, 2=GAMMA
    size_dist = 0
!*** Fit size distribution parameter reff: 0=no, 1=yes
    reff_fit = 1
!*** (Initial) size distribution parameter reff
    reff = 4.
!*** Fit size distribution parameter veff: 0=no, 1=yes
    veff_fit = 0
!*** (Initial) size distribution parameter veff
    veff = 10.
!*** Fraction of spherical particles for aerosol, fraction of plate particles for
cirrus
    shapefrac = 1.
!*** Fit shapefrac: 0=no, 1=yes
    shapefrac_fit = 0
!*** Tilted angle (only relevant for cirrus)
    tilt_angle = 30. /

```